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SLIATE

SLIATE
SRI LANKA INSTITUTE OF ADVANCED TECHNOLOGICAL EDUCATION

(Established in the Ministry of Higher Education, vide in Act No. 29 of 1995)

Higher National Diploma in Information Technology
Second Year, First Semester Examination – 2021
HNDIT 2302 IT Project Management

Instructions for Candidates:
Answer any **04 Questions**
All questions carry equal marks.
Only non-programmable calculators are allowed.

No. of questions : 05
No. of pages : 03
Time : Two (02) hours

Q1)

i. What is meant by **Project**? (03 Marks)

ii. Write four (04) attributes of project. (04 Marks)

iii. "The **Triple Constraint** of project is very important factor in Project Management". What are those Triple Constraints and Why is it important to the project? (05 Marks)

iv. Who are the **Project Stakeholders**? Give two examples. (05 Marks)

v. "Project Management knowledge areas are important components in Project Management Framework". What are those knowledge areas? (08 Marks)

[Total 25 Marks]

Q2)

i. What is meant by "**Strategic Planning**" in Project Integration Management? (03 Marks)

ii. List four (04) methods of selecting a project? (04 Marks)

iii. "A Project Charter is a document that formally recognizes the existence of a project and provides direction on the project's objectives and management." Write 05 components Project Charter should include? (05 Marks)

iv. What is meant by Payback Period Analysis? List two (02) advantages and two (02) disadvantages on Payback Period Analysis? (05 Marks)

v. Nihal Group of Finance have an investment having initial cash outflow of Rs. 250,000. The cash inflows during the first, second, third and fourth years are expected to be Rs. 66,000, Rs. 78,000, Rs. 92,000 and Rs. 105,000 respectively.

a) Find NPV (Net Present Value) for the rate of 10% and 13% (05 Marks)

b) What is IRR and calculate IRR for above project. (03 Marks)

[Total 25 Marks]

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Q3)

i. What are the main phases in Project Time Management Process? (03 Marks)

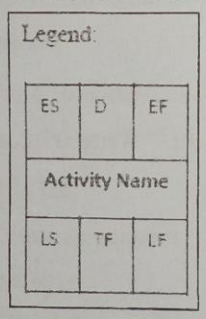
ii. Differentiate between PDM (Precedence Diagramming Methods) and ADM (Arrow Diagramming Methods) diagramming methods? (04 Marks)

iii. Assume that you are the project manager in IT department of a Commercial bank. The bank is going to provide transaction facilities over Internet and ATM machines. In this task, you have prepared an outline of main activities as follows.

Activity	Description	Duration (in weeks)	Predecessor/Preceded by
A	Prepare the architectural design	5	-
B	Analyze software requirements	8	A
C	Design software modules	3	B
D	Analyze hardware requirements and purchase hardware	4	A
E	Develop and test ATM module	5	C
F	Develop and test web module	7	C
G	Install ATM module on ATM machines	6	D, E
H	Host the web module	8	F
I	Setup ATM machines at branches	6	G
J	Merge with the banking system	5	H, I

a) Draw the Precedence Diagram (PDM)/ Activity on Node (AON) network diagram.

(09 Marks)



ES - Early Start
EF – Early Finish
D – Duration
LS – Late start
LF – Late Finish
TF – Total Float

a) Find the Critical Path. (04 Marks)

b) Find project duration. (02 Marks)

c) Will it affect to the project duration if the activity 'F' takes 8 days to complete? (02 Marks)

d) What is the minimum duration needed to complete the project? (01 Mark)

[Total 25 Marks]

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Q4)

i. List three phases in Project Quality Management Process? (03 Marks)

ii. Testing should be done during almost every phase of the IT product development life cycle. Mention two (02) types of testing use in IT product development? (04 Marks)

iii. What are three (03) basic tools and techniques for Cost Estimates? (05 Marks)

iv. According to COCOMO (Constructive Cost Estimation Model) was proposed by Boehm, any software development project can be classified into three (03) types. What are those types? (05 Marks)

v. The size of a development project is 100 Kilo Line of Codes and it has been evaluated as simple. Find the value for average staff and productivity using basic COCOMO model. (hint: $E = 2.4 \times (size)^{1.05}$ $TDEV = 2.5 \times (E)^{0.38}$) (08 Marks)

[Total 25 Marks]

Q5)

i. What is meant by "**Project Communication Management**"? (03 Marks)

ii. What is meant by "**Procurement**"? (04 Marks)

iii. What are processes of Risk Management? (05 Marks)

iv. Which kind of skills require to develop for project managers? (05 Marks)

v. Briefly explain the following:

a. Statement of Work (SOW)

b. Collaborative Procurement

c. Decision Trees

d. Expected Monetary Value (EMV) (08 Marks)

[Total 25 Marks]

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